

## Year 7 • Chapter 1 • Integers: answers (teacher copy)

### 1.01 Integers and opposites (booklet p. 3)

#### WE DO examples

- Ex 1:  $-6$ ;  $+4$
- Ex 2:  $9$ ,  $-15$ ,  $0$
- Ex 3:  $-3$ ,  $+3$ ; back where it started
- Ex 4:  $-8$  (8 units from 0)
- Ex 5:  $-25$
- Ex 6:  $7$  and  $-7$

#### Set A

- $-5$
- $40$
- $-12$
- $8$
- $-3$
- $200$

#### Set B

- $2$
- $23$
- $-56$
- $10$
- $-8$
- $0$

#### Set C

- 3 steps back;  $+3$  and  $-3$
- $+14$  m (14 m above sea level)
- $-7$
- $-12$  (12 units from 0)
- $5$  and  $-5$
- You owe \$60 (overdrawn)

#### Set D

- No,  $-0 = 0$
- $-18$
- $-2$ ,  $-1$ ,  $0$ ,  $1$ ,  $2$
- Yes: every integer has an opposite (0 is its own)

#### Extension

- $11$ ;  $-11$
- No: if  $n$  is negative,  $-n$  is positive (e.g.  $n = -3$ ,  $-n = 3$ )

#### Exit ticket

- Level 1:  $15$
- Level 2:  $-9$ ;  $9$
- Level 3:  $31$ , because the opposite of  $31$  is  $-31$

### 1.02 The number line (booklet p. 7)

#### WE DO examples

- Ex 1: dots at  $-5$ ,  $-2$ ,  $0$ ,  $3$
- Ex 2:  $-2$ ,  $-1$ ,  $0$ ,  $1$
- Ex 3: dots at  $-4$ ,  $-1$ ,  $2$ ,  $5$
- Ex 4:  $-5$
- Ex 5: scale in 5s; dots at  $-30$ ,  $-15$ ,  $10$
- Ex 6:  $9$

#### Set A

- $-7$
- $-3$
- $0$
- $2$
- $5$
- $-1$

#### Set B

- dots at  $-3$ ,  $-1$ ,  $1$ ,  $5$ ,  $6$
- dots at  $-7$ ,  $-4$ ,  $0$ ,  $4$

#### Set C

- $-1$ ,  $0$ ,  $1$ ,  $2$ ,  $3$ ,  $4$
- $-5$ ,  $-4$ ,  $-3$ ,  $-2$ ,  $-1$ ,  $0$
- $-7$ ,  $-6$ ,  $-5$ ,  $-4$
- $-6$ ,  $-4$ ,  $-2$ ,  $0$ ,  $2$ ,  $4$ ,  $6$

#### Set D

- scale  $-40$  to  $40$  in 10s with dots
- $-3$ ,  $-2$ ,  $-1$ ,  $0$ ,  $1$ ,  $2$
- $17$
- $-3$

#### Extension

- $54$
- $-6$

#### Exit ticket

- Level 1:  $-2$ ,  $-1$ ,  $0$
- Level 2:  $-8$
- Level 3:  $-3$

### 1.03 Comparing and ordering integers

(booklet p. 11)

#### WE DO examples

- Ex 1: False; false
- Ex 2:  $-6$ ,  $-1$ ,  $0$ ,  $3$ ,  $5$
- Ex 3:  $>$
- Ex 4:  $8$ ,  $3$ ,  $0$ ,  $-2$ ,  $-15$
- Ex 5:  $-5$ ,  $-(2)$ ,  $1$ ,  $-(-3)$
- Ex 6:  $-11$ ,  $-10$ ,  $-9$

#### Set A

- F
- F
- T
- T
- T
- F

#### Set B

- $>$
- $<$
- $>$
- $>$
- $<$
- $<$

#### Set C

- $-8$ ,  $-2$ ,  $0$ ,  $4$ ,  $7$
- $14$ ,  $3$ ,  $0$ ,  $-1$ ,  $-5$ ,  $-9$ ,  $-10$
- $-20$ ,  $-15$ ,  $-3$ ,  $8$ ,  $12$
- $6$ ,  $-1$ ,  $-2$ ,  $-7$ ,  $-11$
- $-12$ ,  $-4$ ,  $-1$ ,  $0$ ,  $3$
- $40$ ,  $12$ ,  $-8$ ,  $-35$

#### Set D

- e.g.  $-6$ ,  $-7$ ,  $-8$
- $-8$
- $-1000$ ,  $-100$ ,  $-10$ ,  $-1$ ,  $0$
- Yes, for negatives:  $-20$  is smaller than  $-2$

### Extension

1.  $-4, -(-2), 3, -(-4), -(-9)$
2. e.g. 0 and 2 (any integer from  $-3$  to 2)

### Exit ticket

1. Level 1: False
2. Level 2: 2, 0,  $-1, -5$
3. Level 3:  $-20, -12, 0, 5, -(-8)$

## 1.04 Adding integers (booklet p. 15)

### WE DO examples

1. Ex 1:  $-1$
2. Ex 2:  $-10; 4$
3. Ex 3:  $-11$
4. Ex 4:  $1$
5. Ex 5:  $5^{\circ}\text{C}$
6. Ex 6:  $-13$

### Set A

1. 3
2. 4
3.  $-7$
4.  $-3$
5.  $-10$
6. 3

### Set B

1.  $-5$
2. 0
3.  $-15$
4. 14
5.  $-5$
6.  $-14$

### Set C

1.  $-11$
2. 1
3.  $-8$
4. 4
5.  $-11$
6.  $-10$

### Set D

1.  $5^{\circ}\text{C}$
2. level 5
3.  $-\$15$
4. 6

### Extension

1.  $-12$
2. 7

### Exit ticket

1. Level 1: 3
2. Level 2:  $-7$
3. Level 3:  $1^{\circ}\text{C}$

## 1.05 Subtracting integers (booklet p. 19)

### WE DO examples

1. Ex 1:  $-4; -12$
2. Ex 2:  $7; 4$
3. Ex 3:  $-12$
4. Ex 4: 6
5. Ex 5:  $15^{\circ}\text{C}$
6. Ex 6:  $-8$

### Set A

1.  $-4$
2.  $-12$
3.  $-7$
4.  $-7$
5.  $-5$
6.  $-20$

### Set B

1. 7
2. 4
3. 0
4. 10
5.  $-9$
6. 7

### Set C

1.  $-12$
2.  $-7$
3. 6
4.  $-11$
5.  $-5$
6. 0

### Set D

1.  $16^{\circ}\text{C}$
2. 25 m
3.  $-75$  m (75 m deep)
4.  $-13$

### Extension

1.  $-8$
2.  $-10$

### Exit ticket

1. Level 1:  $-9$
2. Level 2: 13
3. Level 3:  $19^{\circ}\text{C}$

## 1.06 Multiplying integers (booklet p. 23)

### WE DO examples

1. Ex 1:  $-18; -10$
2. Ex 2:  $48; -24$
3. Ex 3:  $-42$
4. Ex 4: 40
5. Ex 5:  $-8$
6. Ex 6: Positive: 100 negatives make 50 pairs

### Set A

1.  $-18$
2.  $-10$
3.  $-44$
4.  $-63$
5.  $-24$
6.  $-100$

### Set B

1. 48
2. 25
3. 21
4. 24
5. 20
6. 36

**Set C**

1. -24
2. -42
3. 40
4. 9
5. 100
6. -8

**Set D**

1. -28 m (28 m deep)
2. -7
3. Negative: 2 pairs make positive, 1 negative left
4. -18°C

**Extension**

1. -32
2. 8 and -3

**Exit ticket**

1. Level 1: -42
2. Level 2: 72
3. Level 3: -60

**1.07 Dividing integers** (booklet p. 27)**WE DO examples**

1. Ex 1: 3; -3
2. Ex 2: 2
3. Ex 3: -3
4. Ex 4: 8
5. Ex 5: -\$12
6. Ex 6: 54

**Set A**

1. 3
2. -3
3. 1
4. -4
5. -2
6. -100

**Set B**

1. -5
2. 7
3. -9
4. 4
5. 0
6. -9

**Set C**

1. 2
2. -3
3. 8
4. -4
5. 2
6. 8

**Set D**

1. -\$21
2. -3°C each hour
3. -35
4. -8

**Extension**

1. -60
2. -5

**Exit ticket**

1. Level 1: -4
2. Level 2: 7
3. Level 3: -4

**1.08 Order of operations with integers**

(booklet p. 31)

**WE DO examples**

1. Ex 1: -12
2. Ex 2: -4
3. Ex 3: -18
4. Ex 4: 7
5. Ex 5:  $1 \div (-3) = -1/3$
6. Ex 6: 0

**Set A**

1. -14
2. -13
3. 0
4. 13
5. 18
6. -3

**Set B**

1. -18
2. -19
3. -15
4. -2
5. 15
6. 2

**Set C**

1. 7
2. 0
3. 12
4. 2
5. -1
6. -2

**Set D**

1. -4
2.  $-1/3$
3. 4
4. 4

**Extension**

1.  $(8 - 12) \div (-4 + 2) = 2$
2. e.g.  $-3 \times 4 - 5 = -17$

**Exit ticket**

1. Level 1: 1
2. Level 2: 15
3. Level 3: 12

**1.09 Integers on a calculator** (booklet p. 35)**WE DO examples**

1. Ex 1: -9; -6
2. Ex 2: 5
3. Ex 3: -12
4. Ex 4: 87
5. Ex 5: -52
6. Ex 6: 4

**Set A**

1. -9
2. -6
3. -83
4. -48
5. 12
6. -75

**Set B**

1. -12
2. -52
3. 87
4. -4
5. -7
6. 169

**Set C**

1. 5
2. 9
3. 36
4. 35
5. -3
6. 36

**Set D**

1.  $\approx -1000$ ; -1029
2.  $\approx -20$ ; -20.5
3. The calculator squares 6 first, then makes it negative.  
Enter  $(-6)^2$
4. It should be 8:  $\times$  is done before  $-$

**Extension**

1. 5
2. -5

**Exit ticket**

1. Level 1: -45
2. Level 2: 5
3. Level 3: 5

**1.10 Integer problems** (booklet p. 39)**WE DO examples**

1. Ex 1:  $16^{\circ}\text{C}$
2. Ex 2:  $-\$45$
3. Ex 3: level -3
4. Ex 4:  $\$85$
5. Ex 5:  $\$820$
6. Ex 6: -12 m

**Set A**

1.  $17^{\circ}\text{C}$
2.  $-4^{\circ}\text{C}$
3.  $-3^{\circ}\text{C}$
4.  $-2^{\circ}\text{C}$

**Set B**

1. 12 m deep (-11 m)
2. 5 floors
3. 550 m
4. 155 m

**Set C**

1.  $-\$25$
2.  $\$45$
3.  $-\$60$
4.  $\$150$

**Set D**

1.  $26 \times 80 - 6 \times 210 = \$820$
2.  $35 - 27 = 8$

**Extension**

1. 70 minutes
2. e.g. It was  $4^{\circ}\text{C}$  and fell  $12^{\circ}\text{C}$

**Exit ticket**

1. Level 1:  $4^{\circ}\text{C}$
2. Level 2: 18 m
3. Level 3: -7